June 16, 2025

The Honorable Mehmet Oz
Administrator
Centers for Medicare & Medicaid Services
U.S. Department of Health and Human Services
200 Independence Avenue, SW
Washington, DC 20201

Thomas Keane
Assistant Secretary for Technology Policy
National Coordinator for Health Information Technology
U.S. Department of Health and Human Services
330 C Street, SW
Washington, DC 20201

Re: Sword Health Response to CMS-0042-NC Request for Information: Health Technology Ecosystem

Dear Administrator Oz and Assistant Secretary Keane,

Thank you for the opportunity to provide comments in response to the Health Technology Ecosystem Request for Information (RFI). Enabled by its artificial intelligence (AI) care platform, Sword Health Technologies, Inc. (Sword) delivers world-class healthcare available anytime, anywhere while significantly reducing costs for payers, self-insured employers, and other healthcare organizations. Sword began by reinventing care musculoskeletal (MSK) pain, the most prevalent and costly condition in healthcare, with AI and has since expanded into pelvic health, movement health, and this month, mental health. Since 2020, more than 500,000 members have completed 6.5 million sessions, helping Sword's 1000 enterprise clients avoid nearly $1 billion in unnecessary healthcare costs.

We offer our perspective from our experience delivering AI enabled care. We have focused our comments on where we see AI enabled care delivering higher quality results at a lower cost and where there are challenges or roadblocks preventing expansion of this kind of care for Medicare beneficiaries.

The Promise of AI Enabled Care

Over the last 40 years, nearly every industry except healthcare has seen a massive increase in productivity driven by technological advances. Take consumer electronics as an example: a TV that once cost $3,500 now delivers vastly better performance for $500. Healthcare, by contrast, has experienced only minimal productivity gains over the same period. In healthcare, technology

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is generally only added as a tool for the clinicians, adding a tech layer on top of an already labor-intensive system. Care is still dependent on humans, and access remains limited.

AI enabled care allows us to finally shift labor from humans to machines as other industries did decades ago. This shift unlocks access to high-quality care at scale, making it as seamless and ubiquitous as running water. It also drives down cost, because when patients get the right care early, they avoid the maze of potentially ineffective and expensive interventions that define today's system.

This is the true promise of AI in healthcare: it can be the great equalizer, ensuring that everyone has access to the same high-quality care. First, AI enabled care facilitates a radically better patient experience. People can recover where they are most comfortable: at home. Care becomes convenient and flexible, available instantly when the patient needs it, not when a doctor's office happens to be open.

Second, AI enabled care improves access to high-quality care by removing unnecessary barriers – no two-hour drive to the right office, no six-week wait to get an appointment, no inconvenient appointment times that make patients decide between working and receiving the care they need. AI reduces the friction presented by our current healthcare system meaning patients will get the care they need when they need it, driving better clinical outcomes. For an MSK patient, this means getting early and appropriate treatment for their back pain. This helps the patient avoid unnecessary visits to the emergency room (ER) or more costly procedures, which can result in the prescription of an opioid drug or unnecessary downstream services. These ER visits increase the risk of unnecessary spine surgeries, up to 50% of which are potentially avoidable. For a pelvic health patient, this means early intervention for incontinence, focusing on pelvic floor muscle training, avoiding years of suffering, limitations in day-to-day life, and unnecessary sling surgeries.

Lastly, with improved outcomes and less reliance on unnecessary, wasteful interventions, we finally achieve what healthcare has long needed: a meaningful reduction in costs. In short, AI is the key to bending the cost curve, by delivering better care, to more people, at a fraction of today's cost.

Below we offer our perspective from our experience delivering AI enabled care. Overall, we appreciate that the Centers for Medicare & Medicaid Services (CMS) has identified and defined "digital tools" and "digital health products." The Sword platform and tools squarely fall within these definitions, and we appreciate CMS's consideration of the market and landscape of these products for Medicare beneficiaries. We have focused on our comments to share where we see AI delivering higher quality results at a lower cost and where there are challenges or roadblocks preventing expansion of this kind of care into the Medicare program.

PC-1: What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

Since its inception, Sword set out to change care delivery entirely, because people deserve more than temporary fixes, pills, and waiting rooms. They deserve lasting recovery, powered by technology and grounded in human care. Sword enables convenient, flexible, and instantly available high-quality care at home, improving patient experience and clinical outcomes. This breakthrough in care is only possible with our AI delivery platform, Phoenix, a Food and Drug Administration (FDA)-listed Class II medical device technology. Phoenix delivers real time biofeedback (not just tracking data), enabling Phoenix to act as a therapy assistant, guiding patients through their recovery and providing feedback to the clinicians that oversee the patient's care. Our AI delivery platform and device technology also facilitates task like personalized exercise programs, progress tracking, and access to support, all from the comfort of the individual's home.

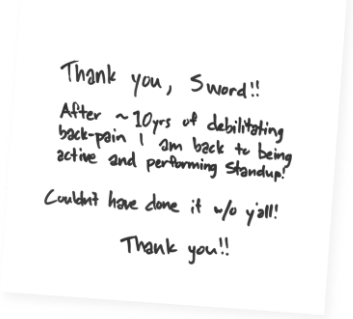
Patients benefit from care delivered through AI platforms, as demonstrated by our clinical outcomes and member satisfaction. In a 12-month period, 67% of our patients report being pain free; 64% report a decrease in feelings of depression; and 70% report a decrease in surgical intent.

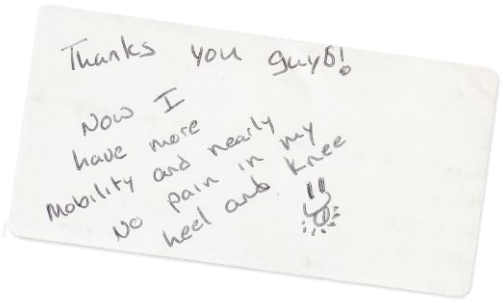
The quality of care that can be delivered by Sword's AI delivery platform is validated by over 40 clinical peer-reviewed papers (available upon request). Through these, we have demonstrated that we can:

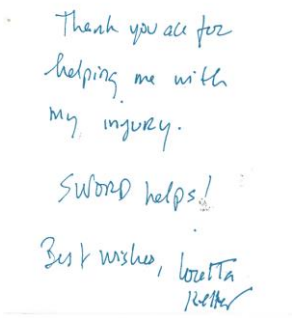
1. Attain better and faster clinical outcomes compared to high-intensity, in-person physical therapy (PT) in the recovery after major MSK surgery;
2. Attain similar clinical outcomes but much higher completion rates than high-intensity, in person PT in chronic MSK complaints;
3. Deliver similar (and significant) clinical and productivity improvements in:
    a. Individuals residing in rural versus urban areas;
    b. Individuals from different racial and ethnic backgrounds (contradicting previous evidence that some groups respond less well to treatment); and
    c. Individuals residing in socially deprived areas, versus those living in less deprived areas (showing our service delivery model can benefit patients equally if given access to care, despite a higher disease burden);
4. Deliver significant improvements in pain, disability, anxiety, depression, as well as productivity, in women suffering from urinary incontinence and those in the menopause stage; and
5. Apply AI to safely scale care, allowing PTs to treat more members while preserving clinical outcomes.

Evidence does not tell the whole story of the impact though. We capture the anecdotal as well. Patients often send us handwritten notes when returning our FDA-listed devices, unsolicited.

These notes tell us what the evidence cannot – AI delivery platforms like Sword can change patients' lives for the better.







All Americans, including Medicare beneficiaries, should have access to new, innovative, and clinically effective tools like Phoenix. Accordingly, Sword encourages CMS to ensure that there are appropriate coverage and reimbursement pathways to enable greater innovation and support adoption, thereby promoting access, to proven services like Sword's AI delivery platform.

PC-5: What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?
PR-1 What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

At Sword, we have replaced the traditional, fully labor-intensive model – where accessing care means waiting for a physical therapist, pelvic health specialist, or psychologist – with a model where patients recover at home, guided by our AI care specialist. Clinicians remain in charge of the care providing oversight and the human touch when it matters most. AI handles everything that does not require a human, so clinicians can focus on what they do best: connecting with patients, ensuring the care is effective, and delivering empathy.

Today, healthcare regulations focus decision making and rules based on where the provider is located, often assuming a hospital or doctor's office is the epicenter of where costs are incurred and care is provided. However, patients need a system that recognizes that AI enabled care can deliver care where the patient is located when the patient needs it. We encourage CMS to consider that costs can also be incurred digitally or where the patient is located and to consider that AI enabled care models are delivered mainly asynchronously, which has implications on how such care should be reimbursed. Only by recognizing these differences, can CMS modify and modernize its approach to the future landscape of health care delivery, ultimately adopting coverage and payment policies reflecting these changes.

Our efforts have been critically important to delivering high quality care to rural patients. (PR-1). In 2018, about three-fourths of Medicare fee-for-service (FFS) beneficiaries lived in full or partial

medically underserved areas (MUAs), with 18%in full MUAs and 60% in partial MUAs as reported in the Medicare Payment Advisory Commission's (MedPAC's) June 2022 *Report to Congress: Medicare and the Health Care Delivery System*.[1] Rural patients, in our experience, face multiple barriers including lack of access to care centers, qualified clinicians, and reliable Wi-Fi. Sword's offerings enabled through asynchronous AI enabled care means that a rural patient can do their treatments in their own home without a multi-hour drive to and from a physical therapist's office. For those without reliable Wi-Fi, Sword can provide a hotspot to enable real time biofeedback from our FDA-listed device and AI enabled care system. When healthcare approaches the patient with an eye to removing barriers, results follow. Our clinical studies have shown high acceptance and satisfaction with such care models in individuals living in rural areas, with higher program completion rates than their counterparts living in urban areas.

This is key to unlocking the promise of AI – enabling care to be delivered without barriers and ensuring access is not determined based on where a patient lives. We encourage CMS to consider all the needs of patients when considering how patients access high quality care, as well as ways CMS can and should support access in rural and underserved areas. As we have found in our experience, Sword can deliver better care while avoiding unnecessary costs, and this should be made available to Medicare beneficiaries.

PC-6: What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

For our patients at Sword, providing an FDA-listed device that provides real-time feedback has been a boon for our customers with limited experience with digital tools and services. Our FDA-listed device, the Thrive tablet, is a 10-inch high-definition (HD) screen with enhanced noise cancellation and improved voice communication. This allows the patient to receive improved biofeedback with the latest technology without impacting patients who may not consistently have access to a smartphone or a smartphone with adequate technology. Bringing the best of user experience and interface design principles to health care also means that Sword meets our patients where they are in terms of their technological advancement. We encourage CMS to consider how these types of devices improve the quality of care when it considers coding, coverage, and payment of these medical devices. At Sword, these investments in technology have enabled high quality and lower cost care for all, not just for those savvy with technology and with the means to purchase the latest smartphone.

[1] June 2022 Report to the Congress: Medicare and the Health Care Delivery System, Medicare Payment Advisory Commission (June 15, 2022), https://www.medpac.gov/document/june-2022-report-to-the-congress-medicare-and-the-health-care-delivery-system/.

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PC-7: If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

Sword believes that AI enabled care delivers better results and avoids downstream costs. To ensure Medicare beneficiaries can access innovative care, like Sword's offerings, and to better support the sustainability and longevity of the Medicare program, we encourage CMS to consider the added benefit of these products and ways to account for the improved clinical outcomes they provide. In short, we recommend that CMS consider how it values and supports adoption of innovative and proven technological advancements and update policies to ensure Medicare beneficiaries have access to these digital health products.

Sword has extensive experience with various payment arrangements in the private sector, and most specifically with outcome-based payment models. For example, payers, self-insured employers and other healthcare organizations pay the costs of Sword's AI enabled platform for their members based on outcomes. Sword uses the Patient Global Impression of Change (PGIC) questionnaire, which is a consumer-friendly, single-item, seven-point scale used to assess the patient's perception of their overall improvement following treatment. This scale is widely used in clinical trials, especially in pain research to capture a patient's experience of change. In the Sword pricing model, if a patient reports a successful outcome based on their PGIC score, then Sword bills the payer for the full amount of the treatment.

Based on our experience, savings from outcome-based contracts can fund $0 cost-sharing and targeted roll-outs for example in rural or dually-eligible populations, where the benefits of digital health products might be most beneficial. CMS can modernize digital health reimbursement by moving from "minutes monitored" to "outcomes delivered." An outcomes-based episode, grounded in the real-world data infrastructure outlined above, will (a) protect beneficiaries from ineffective products, (b) unleash innovation among high-performing vendors, and (c) curb the unchecked spending growth from the past few years.

TD-1: What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?
VB-1: What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients? What are the current obstacles preventing broader digital product adoption for patients in ACOs?
VB-2 How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

Sword's AI enabled care demonstrates that there is an effective way to manage chronic diseases, like MSK conditions, cost-effectively and through a non-invasive, non-pharmacological approach. Sword's AI enabled approach is not only possible but also demonstrates results for non-Medicare populations today. Through our AI enabled care platform, patients receive high quality care with over 85% of patients experiencing clinically meaningful changes as measured through the PGIC score. These patients also note clinically significant improvements in pain, disability, productivity and mental health, with data from our publications showing 55-64% reduction in pain, 68-79% recovery in productivity impairments, 50-54% reduction in anxiety scores, and 50-65% reduction in depression scores.

Our AI enabled care also allows us to tackle wasteful spending, helping eliminate unnecessary or non-evidence-based care. In MSK care, regional variations in surgery rates reach 400% according to Dartmouth Atlas.[2] Our own studies show that approximately 36% of MSK surgeries are unnecessary and potentially avoidable,[3] a number which can increase to 50% in the case of spinal surgery according to third party studies.[4]

Because our AI enabled care provides the right treatment at the right time, payers, self-insured employers, and other healthcare providers who utilize Sword have seen a reduction in unnecessary healthcare costs. Overall, every dollar invested in Sword returns $3.20 in savings (independently validated; available upon request). More specifically, a top health plan saved $25.1 million in unnecessary health care costs, resulting in a 2.7x saving compared to the investment in Sword, as calculated by the health plan's actuaries. A Fortune 50 consumer company achieved 3.7x savings on their population, and a group of clients from different industries achieved an independently validated 3.2x saving on their population.

AI enabled care makes our clinicians more effective and efficient, focusing on patient care rather than administrative duties, and enabling remote care delivery. This is particularly critical given expected physical therapist shortages that the industry is facing, particularly in rural and underserved areas where specialist shortages are most acute.[5]

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[2] See Goodney PR, Dzebisashvili N, Goodman DC, Bronner KK. Variation in the Care of Surgical Conditions. 2015, https://data.dartmouthatlas.org/downloads/atlases/Surgical_Atlas_2014.pdf.

[3] See The MSK Money Pit: Where Benefit Budgets Go To Die. Sword Health, March 13, 2024, https://swordhealth.com/insights/gated-reports/msk-money-pit.

[4] See Vanamadala V, Kim Y, Buchlak QD, Wright AK, Babington J, Friedman A, Mecklenburg RS, Farrokhi F, Leveque JC, Sethi RK. Multidisciplinary Evaluation Leads to the Decreased Utilization of Lumbar Spine Fusion: An Observational Cohort Pilot Study. Spine (Phila Pa 1976). 2017 Sep 1;42(17):E1016-E1023. doi: 10.1097/BRS.0000000000002065. PMID: 28067696.).

[5] See Patrick Zarek, Clark Ruttinger, David Armstrong, Ritashree Chakrabarti, Douglas R Hess, Tara Jo Manal, Timothy M Dall, Current and Projected Future Supply and Demand for Physical Therapists From 2022 to 2037: A New Approach Using Microsimulation, Physical Therapy, Volume 105, Issue 3, March 2025, pzaf014, https://doi.org/10.1093/ptj/pzaf014.

To stimulate interest across developers and value-based care entities, CMS must take steps to modernize its approach to digital health products, especially under Medicare FFS. By doing so, CMS can establish appropriate coverage and payment, giving comfort to companies seeking to innovate and healthcare providers seeking to improve outcomes, that support the availability and adoption of AI enabled care like Sword's platform. The current fragmented Medicare FFS approach is workable, but CMS should take steps to ensure it can meaningfully value and support adoption of digital health products under this framework. Concurrently, CMS should ensure that ACOs and MSSP participants are fully aware of these product offerings and be encouraged to adopt and use them. CMS should also ensure that appropriate guardrails exist to limit unnecessary utilization and support access to proven digital health products like Sword's offerings.

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Thank you for the opportunity to respond to this RFI. We appreciate CMS's focus and attention on digital tools and digital health products. We share CMS's goal of promoting access to innovative care and protecting Medicare beneficiaries and the program by ensuring these products are appropriate and effective. We believe Sword's AI enabled care meets all of CMS's goals and we look forward to the opportunity to work with you. Please feel free to contact Beth Stevens, Chief Legal Officer at beth@swordhealth.com should you have any questions or if you would like to discuss further.

Sincerely,

Virgílio Bento
Chief Executive Officer